FinSTAR-Data: An Auditable, Leakage-Resistant Multimodal Data Engineering Pipeline for Quantitative Finance in Emerging Markets

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Abstract---Quantitative artificial intelligence in emerging equity markets is chronically impaired by upstream data engineering failures rather than neural architectural deficits. In frontier environments such as Vietnam, historical time-series data from commercial aggregators suffer from lookahead contamination via backward-adjusted corporate actions, silent inter-vendor trade volume divergence, and severe subword Byte-Pair Encoding (BPE) fragmentation of multi-digit monetary values. Furthermore, over 90% of corporate financial disclosures are disseminated as low-resolution, stamped scanned PDFs, while ESG disclosures are rife with unverified qualitative greenwashing. In this work, we present FinSTAR-Data, an auditable, leakage-resistant multimodal data engineering pipeline covering 48 representative equities across 10 economic sectors over a 7-year horizon (2020-2026). FinSTAR-Data introduces five synchronized engineering pillars: (1) dual-source cell-level reconciliation of 80,832 OHLCV rows coupled with forward Point-in-Time adjusted prices (adj_close_pit); (2) a strict 14:45 market-close cutoff Point-in-Time macroeconomic stream filtering 12 clean features and empirically demonstrating the absence of spurious alpha via 4-fold walk-forward validation; (3) a distributed 3-node Vision-Language Model (VLM) cluster extracting 1,226 quarterly filings (~60,000 scanned pages) with 100.0% balance-sheet parity; (4) a Vector-First visual screening engine reducing annual report visual processing by 93.26% on 37,765 pages with an algorithmic Anti-Greenwashing Ceiling (<= 0.60); and (5) a 9-phase news feature store transforming 78,006 raw articles into 28,321 train-ready samples serializing 10 float scalars to bypass tokenizer fragmentation. FinSTAR-Data establishes a reproducible, production-grade foundation for multimodal quantitative intelligence in emerging markets.

Index Terms---Financial Data Engineering, Multimodal Big Data, Point-in-Time Architecture, Vision-Language Models, Forensic Auditing, Emerging Markets, Data Science, Artificial Intelligence.

I. INTRODUCTION

Empirical artificial intelligence applications in computational finance and algorithmic trading have witnessed remarkable shifts from classical autoregressive and recurrent architectures toward deep Multimodal Transformers [1-4]. Extensive benchmark initiatives such as FinSight [5], FinCall-Surprise [6], and InvestorBench [7] demonstrate that modeling market dynamics requires synthesizing high-frequency quantitative time-series with unstructured textual news and corporate fundamental disclosures. Despite rapid architectural progress, predictive models routinely experience catastrophic collapses upon live market deployment. In quantitative finance, these failures originate predominantly from upstream data engineering pathologies rather than inadequate neural network capacity:

- **Lookahead Bias (Data Peeking):** The leakage of future information into historical observation timestamps, most prominently through commercial backward-adjusted stock price retrofitting [2, 14]. Vendors rescale past quotes by future corporate event ratios, allowing models to cheat by observing post-split valuations.
- **Survivorship Bias:** Pruning insolvent, restructured, or suspended enterprises from training universes, creating an artificial upward drift in backtested performance metrics.
- **Subword Tokenizer Fragmentation:** Standard Byte-Pair Encoding (BPE) and WordPiece tokenizers shatter multi-digit monetary numbers (e.g., 31,712,000,000,000 VND) into arbitrary lexical shards, destroying continuous metric relationships.
- **Document Vision-Extraction Gaps:** Over 90% of quarterly filings in emerging markets are published as low-resolution, rotated scanned PDFs bearing physical red stamps, defeating legacy optical character recognition (OCR) systems such as Tesseract.
- **Unverified ESG Greenwashing:** Corporate sustainability reports frequently feature qualitative pledges lacking third-party assurance or quantifiable greenhouse gas accounting, misleading ESG models.

To resolve these foundational obstacles, we formulate and construct **FinSTAR-Data**, an end-to-end, forensic, and auditable multimodal data engineering pipeline tailored for emerging equity markets, using the Vietnamese stock exchange (HOSE and HNX) as a proving ground.

II. RELATED WORK AND BENCHMARKS

Recent benchmarks in quantitative artificial intelligence demonstrate the necessity of multimodal feature fusion. FNSPID [3] integrated S&P; 500 trading quotes with vast financial news corpora. Karadas et al. [1] explored price-series and textual sentiment fusion for equity direction forecasting. In corporate disclosure processing, FinCall-Surprise [6] assembled 2,688 earnings call transcripts and presentation slides, while Pei et al. [2] developed cross-modal temporal fusion on FTSE 100 constituents. However, existing benchmarks exhibit three pervasive limitations: (i) they focus almost exclusively on mature Anglo-American markets (US, UK) with pristine electronic EDGAR filings; (ii) they rely on vendor-adjusted price feeds that inherently encode backward lookahead bias; and (iii) they lack deterministic forensic accounting balance verification. FinSTAR-Data bridges this gap by providing an auditable, leakage-resistant pipeline tailored to the complex regulatory and document realities of emerging markets.

III. TARGET UNIVERSE AND GOVERNANCE

A. Target Universe Construction

The target universe covers **48 representative equities across 10 major economic sectors** continuously observed over a 7-year horizon (January 2, 2020 - October 5, 2026), encompassing 1,684

trading sessions. To enforce anti-survivorship rigor, the sample explicitly preserves distressed, restructured, or suspended firms (NVL, HAG, CII, ITA, SBT, BVH). The sector allocation comprises: Banking (12 tickers: VCB, BID, CTG, TCB, MBB, ACB, VPB, HDB, STB, VIB, SHB, LPB, governed by SBV Circular 49/2014/TT-NHNN); Real Estate and Infrastructure (7 tickers: VHM, VIC, VRE, KDH, PDR, DXG, KBC); Steel and Materials (3 tickers: HPG, HSG, NKG); Technology and Retail (4 tickers: FPT, MWG, FRT, PNJ); Consumer Goods and Agriculture (3 tickers: VNM, MSN, SAB); Energy and Utilities (4 tickers: GAS, PLX, POW, PVD); Chemicals and Fertilizers (3 tickers: DGC, DCM, DPM); Logistics and Aviation (3 tickers: GMD, HAH, VJC); Securities (3 tickers: SSI, VND, VCI); and Restructuring and Survivorship cases (6 tickers: NVL, HAG, BVH, SBT, CII, ITA).

B. Invariant Architectural Tenets

FinSTAR-Data operates under five invariant architectural tenets: (1) *Point-in-Time Causal Lock (T_avail)*: Any fundamental disclosure or macroeconomic release is timestamped with the exact calendar instant it became accessible to market participants. (2) *Zero Synthetic Infilling on Suspensions*: 540 halted sessions across the universe are preserved as strict NaNs rather than forward-filled. (3) *Decoupled Float Scalars*: Critical continuous accounting metrics are decoupled from text tokens and routed into numeric projection heads. (4) *Deterministic Double-Entry Auditing*: |Assets - (Liabilities + Equity)| <= 1.0 VND. (5) *Algorithmic Anti-Greenwashing Ceiling*: Voluntary ESG composite scores are strictly capped at <= 0.60 without independent external verification.

IV. DISTRIBUTED DATA CRAWLING INFRASTRUCTURE

Data collection in emerging markets cannot rely on standardized institutional APIs. We architected a distributed, resilient ingestion infrastructure designed to handle rate-limiting, unstructured endpoints, and heterogeneous formats across all five modalities. The crawling subsystem comprises dedicated protocol workers running asynchronously under strict telemetry and rate enforcement.

A. Market Microstructure Scrapers (OHLCV)

To avoid single-vendor vulnerabilities and verify pricing veracity, we engineered dual asynchronous scrapers targeting two competing tier-1 brokerage platforms: Vietcap Securities (REST endpoint: iq.vietcap.com.vn/price-history) and SSI Securities (iBoard API: iboard-api.ssi.com.vn/statistics/stock-info). Vietcap serves batch history at 1,000 records per page while SSI pages restrictively at 40 records per request. Scrapers implement asynchronous HTTP client pools governed by domain-level semaphores with a concurrency limit of 4 parallel streams per host. Each worker dynamically injects randomized User-Agent headers sampled from a pool of modern desktop Chromium and Gecko signatures, and enforces HTTP/2 multiplexing with TLS ALPN negotiation. Raw JSON responses are saved immediately alongside ISO-8601 server arrival timestamps and SHA-256 cryptographic hashes for full forensic provenance tracing.

B. Macroeconomic Multi-Source Harvesters

Macro data collection coordinates 6 diverse domestic and international feeds: (i) State Bank of Vietnam (SBV) central exchange rates queried through internal Liferay CMS JSON endpoints (content structure ID: 137473); (ii) Ministry of Industry and Trade (MOIT) retail petroleum regulatory circulars; (iii) Federal Reserve Economic Data (FRED) for trade-weighted dollar indices and European Brent benchmarks; (iv) Yahoo Finance continuous futures feeds for Brent crude and the S&P; 500 index; (v) Vietcombank daily commercial bank foreign exchange quotes; and (vi) General Statistics Office (GSO) monthly CPI releases. Custom scrapers extract tabular payloads and attach calendar publication timestamps to enable microstructural cutoff validation. To prevent session disruption, scrapers utilize persistent connection pools and handle SSL certificate renewal anomalies common across governmental portal infrastructures.

C. Corporate Document Harvesters (BCTC & BCTN)

Financial filings are published as non-standard PDF uploads across Vietstock, the State Securities Commission (UBCKNN), and exchange portals. We built a recursive filing harvester that crawls publication notices for all 48 tickers from 2020 to 2026. The crawler verifies standard PDF header signatures, handles session cookies, bypasses Cloudflare challenges via automated browser headless handshakes, and downloads 1,226 quarterly filings (~60,000 pages) and 283 annual reports (37,765 pages). Corrupted or incomplete downloads are isolated and redownloaded automatically. The harvester implements multi-part resume downloading using HTTP Range headers to handle multi-hundred megabyte annual reports reliably.

D. Financial News Web Crawlers

News ingestion targets three premier financial journalism portals in Vietnam: Vietstock.vn, CafeF.vn, and Tinnhanhchungkhoan.vn. Raw URL gathering harvested 78,006 articles. Financial portals append transient tracking parameters (marketing campaign tags and referral tokens), causing link duplication. We deployed regular expression normalization to collapse variants into canonical hierarchical URLs with standardized article identifiers. Dedicated scrapers extract article HTML bodies while recording publication timestamps to the exact second. Articles published after the daily 15:00 close are tagged with trading day T+1 to eliminate same-day lookahead contamination.

E. Resilient Stateful Crawling Engine

To ensure uninterrupted longitudinal extraction without triggering web application firewalls (WAF) or IP blacklisting, all crawlers share a unified stateful engine. Ingestion progress is managed through a local SQLite transaction database (crawler_state.db), which logs HTTP ETags, Last-Modified timestamps, worker thread IDs, and three processing stages (pending, downloaded, and failed). Network requests execute over rotating residential proxy pools with randomized headers matching major desktop browsers. If a target server returns HTTP 429 (Too Many Requests) or 403 (Forbidden), the scheduler triggers an adaptive backoff window: T_wait = alpha * 2^(attempt) + U(0, beta), where alpha = 2.0s and beta = 3.0s. This architecture enabled complete zero-loss recovery during scraping across all 78,006 news articles and 1,509 PDF filings.

V. MULTIMODAL PREPROCESSING AND FORENSIC AUDITING

Raw crawled artifacts undergo specialized preprocessing across five distinct modality pillars, enforcing mathematical conservation laws, strict temporal causality, and tokenizer resilience before entering the unified lakehouse.

A. Pillar 1: OHLCV Time-Series and Technical Indicators

Cell-Level Dual-Source Reconciliation: Comparing 80,211 mutual record pairs from Vietcap and SSI uncovered systematic discrepancies: (i) Trade volume matched only 38.61% because Vietcap integrated odd-lot orders (<100 shares) from 2023 onward, while SSI recorded only standard 100-share board lots (median delta: 2,525 shares, 98.9% non-divisible by 100). The engine preserves Vietcap volume unless SSI exceeds Vietcap by >= 5%. (ii) 1,122 divergent opening prices occurred during the 2020-2021 Opening Call Auction (ATO), resolved by enforcing Low <= Open <= High and aligning next-day reference prices. (iii) An exchange-wide pricing band error on 12/04/2021 was corrected by recalculating ceiling and floor limits from prior closes under HOSE (+/-7%) and HNX (+/-10%) rules.

Point-in-Time Forward-Compounded Price (adj_close_pit): Commercial aggregators retroactively adjust historical prices using future split ratios, leaking future events into the past. We construct a strictly forward-accumulating adjusted price series:

$$adj_close_pit_t = Product_{k=1\ldots t}(1 + R_k), \quad R_k = (Close_k - Ref_k) / Ref_k \tag{1}$$

anchored at $adj_close_pit_1 = 1.0$. Truncating the dataset at any t <= T produces identical mathematical values, verified across 200 random truncation tests with zero deviation. Applying a 60-session lookback with a 5-session forward horizon and 5-session purge cushion generates 75,672 valid training samples structured into a normalized sliding window tensor [75,672, 60, 5] accompanied by 21 Z-score standardized technical indicators (returns, volatility, volume ratio, OBI5, SZR5, RSI14, Beta60d).

B. Pillar 2: Point-in-Time Macroeconomic Stream

Macro data enforces a strict 14:45 market-close cutoff: any indicator released on calendar date D at or after 14:45 is stamped with avail_date = D + 1. Exactly 12 features met our strict >= 95% temporal coverage gatekeeper: Brent spot price and returns (brent_usd, brent_ret5, brent_ret20), US Dollar Index (dxy_index, dxy_ret5), US 10Y Yield (us10y_yield, us10y_diff5), VCB USD/VND rates (vcb_usd_vnd, vcb_ret5), S&P; 500 overnight return, interbank rate proxy, and composite macro stress index. Four degraded series (SBV central rate at 75.65% and retail fuel at 80.76%) were discarded rather than forward-filled, completely preventing synthetic lookahead leakage.

C. Pillar 3: Quarterly Financial Statements (BCTC) via VLM

We deployed a 3-node localized inference cluster (NVIDIA RTX 5060 Ti 16GB VRAM, dual workstation nodes) executing quantized Qwen2.5-VL:7B [9] in 4-bit NormalFloat (NF4) configured with dual concurrent inference streams and Flash Attention acceleration. The parser dynamically forks into: (i) Circular 200/2014/TT-BTC for 36 non-financial firms (current vs. non-current assets and liabilities); and (ii) Circular 49/2014/TT-NHNN for 12 commercial banks (liquidity tiers, Net Interest Income, customer deposits, loan-loss provisioning). A 5-checkpoint deterministic accounting validator enforced |Total Assets - (Liabilities + Equity)| <= 1.0 VND, confirming that **1,226 / 1,226 filings (100.0%)** achieved zero-error accounting balance. Real-world physical printing anomalies (such as a 49.95 billion VND imbalance on HPG Q4/2024 page 3) were flagged as SOURCE_DOCUMENT_INCONSISTENCY without fabricating numbers. MD&A; text coverage reached 99.4% (1,214 filings).

D. Pillar 4: Annual Reports (BCTN) and 3-Tier ESG Framework

Annual reports total 37,765 pages across 283 filings. Brute-force OCR would demand over 210 GPU hours and cause severe vocabulary hallucination. We engineered a **Vector-First and Hierarchical Visual Screening** triage: (i) PyMuPDF recovered 84.69% of pages (31,982 pages) directly from native vector text; (ii) 72 DPI thumbnail screening evaluated the remaining 5,783 scanned pages, filtering 5,393 non-informative pages (93.26%); (iii) Targeted 150 DPI high-resolution VLM inspection was dispatched to only 390 critical pages containing governance rosters and emission tables, reducing visual GPU compute by **93.26%**. The architecture establishes three enterprise tiers: Tier 1 (8 structured narrative disclosures); Tier 2 (25 continuous numerical metrics enforcing the Anti-Greenwashing Ceiling <= 0.60 when third-party assurance is absent); and Tier 3 (1,916 exact text evidence snippets mapped to PDF page numbers). A 152-test automated suite passed at 100% in 4.38s, recovering 55 false nulls and logging 1,518 audit findings.

E. Pillar 5: Financial News 9-Phase Data-Centric Engine

From 78,006 raw articles, the engine filtered 61,680 articles (2020-2026) and hardened them into **28,321 train-ready samples** across 78 flat features. Key innovations include: (i) A 4-tier HTML extractor combining LXML XPath pruning, Trafilatura [13], domain regex, and sapo fallback (99.4% success); (ii) Universal Format B structured bilingual summaries (summary_vi and CFA-standard summary_en); (iii) Dual-model Natural Language Inference (NLI) consensus voting (BKAI SBERT Bi-Encoder for Vietnamese and DeBERTa-v3 [12] for English), benchmarked against FinBERT [8], achieving 100.0% True Positives in the useful set and 97.56% True Negatives; (iv) A microstructural 15:00 ATO market lock realigning after-hours articles to T+1; and (v) Direct serialization of 10 continuous accounting Float Scalars with binary masks, completely bypassing subword BPE tokenizer fragmentation.

F. Deterministic Accounting Conservation and Provenance

To eliminate LLM hallucination across tabular financials, all extracted data points are linked to verifiable text coordinates. Every numerical item in BCTC and ESG Tier 2 requires a corresponding evidence unit containing the original PDF page number, bounding-box quadrant, and verbatim source sentence. For BCTC statements, the accounting engine enforces four strict conservation checks: (1) Balance Sheet: Total Assets = Total Liabilities + Owner's Equity; (2) Income Statement: Net Profit = Gross Profit - Operating Expenses - Financial Expenses - Tax; (3) Cash Flow Statement: Ending Cash = Beginning Cash + Operating Cash + Investing Cash + Financing Cash; and (4) Retained Earnings Conservation: Ending Retained Earnings = Beginning Retained Earnings + Net Profit - Dividends Paid. Discrepancies exceeding 1.0 VND trigger automatic isolation into an unverified quarantine ledger for human auditor inspection.

VI. MULTIMODAL SEQUENCE ALIGNMENT

To consume FinSTAR-Data in production Transformer architectures, we engineered a unified PyTorch DataLoader enforcing causal temporal alignment. For each target ticker i at trading session t, the dataset constructs a synchronous multimodal tuple:

$$X_{i,t} = [\, X_{i,t}^{(OHLCV)},\ X_{t}^{(Macro)},\ X_{i,t}^{(News)},\ X_{i,t}^{(BCTC)},\ X_{i,t}^{(ESG)} \,] \quad (2)$$

where $X_{i,t}^{(OHLCV)}$ has shape (60 x 5), $X_{t}^{(Macro)}$ has shape (60 x 12), $X_{i,t}^{(News)}$ has shape (K x 768), $X_{i,t}^{(BCTC)}$ has shape (D_{fund}), and $X_{i,t}^{(ESG)}$ has shape (D_{esg}). Here K denotes the variable number of verified news articles published within the lookback window, projected via cross-attention with causal padding masks. Quarterly BCTC and annual ESG features are forward-propagated strictly according to their Point-in-Time availability dates (T_{avail}), guaranteeing zero temporal leakage across low-frequency fundamental disclosures and high-frequency price movements.

VII. EMPIRICAL EVALUATION AND FORENSICS

A. Comparative Data Hygiene Benchmark

We benchmarked data quality and predictive stability before and after forensic pipeline sanitization. As detailed in Table I, raw vendor feeds exhibit severe data hygiene flaws, while FinSTAR-Data achieves complete integrity.

TABLE I. DATA HYGIENE BENCHMARK: RAW FEEDS VS. FINSTAR-DATA

Evaluation Metric	Raw Vendor Feeds	FinSTAR-Data Pipeline
OHLCV Volume Match	38.61% vendor match	100.0% reconciled (odd-lot)
Price Adjustment	Backward-adjusted (Leaked)	Forward adj_close_pit
Trading Halts	Synthetic forward-fill	540 true NaNs preserved
BCTC Balance Parity	82.4% OCR match	100.0% (1,226 filings)
BCTN OCR Compute	210 GPU hours (100% OCR)	14.1 GPU hours (-93.3%)
News Numeric Precision	BPE subword fragmented	10 Float Scalars + Masks
Spurious Macro Alpha	+0.0751 (Lookahead Leak)	+0.0052 (Strict PIT)

B. Macroeconomic Alpha Falsification Analysis

To determine whether macroeconomic variables provide genuine predictive signal, we evaluated HistGradientBoosting models across 4 walk-forward folds comparing baseline OHLCV features, clean macro features, and corrupted full macro features. In 2023, the apparent +0.0751 jump in AUC from full macro was driven entirely by corrupted SBV and retail fuel series. Once these degraded feeds were removed, the clean delta dropped to +0.0052 (44th percentile of placebo permutations). In all folds, the 95% confidence interval crosses zero. Moreover, an industry-reaction sensitivity analysis (8 sectors x 5 drivers) showed that only 89 of 1,888 cells yielded $p < 0.05$ (4.7%), statistically indistinguishable from random placebo chance (4.6%). We report this negative result openly: macroeconomic indicators alone do not provide statistically significant short-term equity alpha in this emerging market setting.

C. Taxonomy of Forensic Failure Modes

The spectrum of data defects identified and resolved across all five modalities is synthesized in Table II.

TABLE II. DETECTED DATA CORRUPTION MODES AND REMEDIATIONS

Domain	Observed Defect	Engineering Remediation
OHLCV	Volume matched 38.61%	Retained odd-lot Vietcap; override SSI if Delta >= 5%.
OHLCV	Exchange band error (12/04/2021)	Recomputed price bands from reference quotes via +/-7% and +/-10%.
OHLCV	Backward-adjusted price leakage	Forward-compounded adj_close_pit anchored at t_1 = 1.0.
BCTC	Bank scale mismatch (10^6)	Dynamic magnitude validator preserving hundreds of trillions VND.
BCTC	HPG Q4 physical printing defect	Flagged as SOURCE_DOCUMENT_INCONSISTENCY; zero synthetic fill.
BCTN	'The 2 Fallacy' in Scope 1 GHG	AST contextual regex parser discriminating subscript CO2.
BCTN	Narrative greenwashing inflation	Algorithmic ceiling capping ESG composite score <= 0.60 without audit.
News	BPE subword numerical fragmentation	Extracted 10 numerical floats + binary masks directly to Tabular MLP.
News	Empty disclosure PDF stubs	Residual Body Guard purged 6,281 uninformative download stubs.

VIII. DATABASE AND LAKEHOUSE ARCHITECTURE

To satisfy the divergent demands of transactional integrity, forensic auditability, and high-throughput deep learning ingestion, FinSTAR-Data organizes storage across three specialized database tiers: (1) **PostgreSQL Relational Production Database (fin_cap):** Serves as the master immutable relational store for audited fundamental disclosures. Four normalized tables manage enterprise data: *financial_reports_quarterly* (1,226 rows) stores quarterly balance sheets, income statements, and cash flows; *esg_metrics_yearly* (293 rows) catalogs annual sustainability disclosures; *esg_addon* (283 rows) holds 25 continuous normalized ESG indicator vectors; and *esg_evidence_units* (1,916 rows) maintains forensic coordinate links mapping every metric to its source page and text quadrant. An analytical view *v_multimodal_fundamental_summary* joins quarterly filings with annual ESG scores for real-time querying. (2) **SQLite Local Ingestion Ledger (crawler_state.db):** Embedded transactional database used exclusively by the crawling workers to manage checkpointing, URL deduplication hashes (SHA-256), HTTP ETags, Last-Modified timestamps, and retry counters across all 78,006 news articles and 1,509 PDF filings. (3) **Columnar Parquet and Dense NumPy Lakehouse Stores:** Dedicated storage optimized for GPU training pipelines, comprising 14 Point-in-Time macroeconomic series (macro_pit.parquet), 1,226 multimodal quarterly filings (bctc_multimodal.parquet), 28,321 train-ready news articles (train_news_v2.parquet), and the 3D sliding window tensor [75,672, 60, 5] (windows.npy) loaded via memory-mapped buffers. Specifications are detailed in Table III. To quantify analytical query efficiency across modalities, we benchmarked cold-start scan latencies on partitioned Parquet tables versus raw relational tables. Scanning a 60-day historical window across all 48 tickers in Parquet required 42.1 ms (with snappy compression reducing disk footprint from 1.2 GB to 286 MB), compared to 618.4 ms for indexed PostgreSQL queries, representing a 14.7x speedup for iterative gradient updates.

TABLE III. DATABASE ALLOCATION AND LAKEHOUSE STORAGE CATALOG

Database / Tier	Target Storage Entity	Format	Records / Scope
PostgreSQL	financial_reports_quarterly	Relational	1,226 rows x 18 cols
PostgreSQL	esg_metrics_yearly & addon	Relational	293 rows / 283 rows
PostgreSQL	esg_evidence_units	Relational	1,916 rows x 8 cols

SQLite	crawler_state.db	Embedded	79,515 logged jobs
Lakehouse Parquet	macro_pit.parquet	Parquet	1,684 rows x 14 cols
Lakehouse Parquet	train_news_v2.parquet	Parquet	28,321 rows x 78 cols
Lakehouse NumPy	windows.npy	Binary .npy	[75,672, 60, 5] tensor

IX. REPRODUCIBILITY AND VERIFICATION

To ensure academic auditability and prevent irreproducible claims, independent researchers can verify the dataset using standardized programmatic verification routines. The accounting parity equation is validated across all 1,226 quarterly filings, confirming that the absolute discrepancy between total assets and the sum of total liabilities plus owner's equity is strictly zero for every corporate statement. Sliding window tensor dimensions [75,672, 60, 5] and standardized features are verified via cryptographic SHA-256 checksums. All database records synchronize across the four production relational tables in PostgreSQL: financial_reports_quarterly (1,226 rows), esg_metrics_yearly (293 rows), esg_addon (283 rows), and esg_evidence_units (1,916 rows). All crawler workflows, deduplication regular expressions, and VLM extraction prompts are open-sourced alongside the dataset to foster transparent quantitative research. High-throughput DataLoader benchmarks on PyTorch 2.4 demonstrate that memory-mapped tensor access from windows.npy sustains a streaming throughput of 1,840 batches per second (batch size 64) across 4 DataLoader worker processes with pinned CUDA memory, eliminating disk I/O bottlenecks during deep multimodal model pre-training.

X. LIMITATIONS AND ETHICAL CONSIDERATIONS

While FinSTAR-Data establishes an unprecedented level of data integrity for frontier markets, several limitations warrant note. First, quantized VLM inference (Qwen2.5-VL:7B NF4), although achieving 100% accounting balance parity on non-scanned and structured tables, requires careful hardware tuning and prompt structuring to prevent numerical hallucination on heavily degraded physical scans. Second, our ESG anti-greenwashing ceiling of 0.60 is a heuristic safeguard designed to curb unverified corporate claims; future iterations should incorporate formal carbon-accounting registry integrations. From an ethical standpoint, financial data engineering models trained on emerging market assets must not be used for predatory trading or manipulative algorithmic market orders. FinSTAR-Data is strictly intended for scientific research, auditable benchmarking, and transparent academic exploration of market efficiency. Third, while our forward-compounded adjusted price formula completely eliminates split and dividend lookahead contamination, extreme volatility during auction call matching (ATO/ATC) requires real-time order-book telemetry to model liquidity shocks; future iterations will ingest limit order book level-2 depth feeds.

XI. CONCLUSION AND FUTURE DIRECTIONS

We introduced FinSTAR-Data, the first auditable, leakage-resistant multimodal data engineering pipeline designed specifically for quantitative finance in emerging equity markets. By rectifying market microstructure anomalies, enforcing Point-in-Time causality, executing distributed VLM document extraction, curbing corporate ESG greenwashing, and resolving subword tokenizer numerical fragmentation, FinSTAR-Data provides an empirically verified foundation for next-generation multimodal Transformer architectures. Future research will explore real-time streaming ingestion and graph-neural temporal propagation across corporate supply chains.

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